

Planner

Equilibrium

Firm's problem:

$$\max_{\mathbf{y}} p_1 y_{11} + p_2 y_{21} \text{ subject to } y_{11} \leq \sqrt{-y_{21}}$$

If $\mathbf{p} \gg \mathbf{0}$, then there is no disposal. Substituting:

$$\max_{y_{21}} p_1 \sqrt{-y_{21}} + p_2 y_{21}$$

First-order condition:

$$-\frac{p_1}{2\sqrt{-y_{21}}} + p_2 = 0$$

Solving for y_{21} :

$$\begin{aligned} p_2 &= \frac{p_1}{2\sqrt{-y_{21}}} \\ \sqrt{-y_{21}} &= \frac{p_1}{2p_2} \\ -y_{21} &= \left(\frac{p_1}{2p_2} \right)^2 \\ y_{21} &= -\frac{p_1^2}{4p_2^2} \end{aligned}$$

And so $y_{11} = \frac{p_1}{2p_2}$.

Note: we know that the planner chooses $y_{11} = \sqrt{8}$, so this means $\frac{p_1}{p_2} = 2\sqrt{8} = \sqrt{4 \times 8} = \sqrt{32}$. This also produces the right labor demand, which is $\frac{32}{4} = 8$.

The profit function is then:

$$p_1 y_{11} + p_2 y_{21} = p_1 \frac{p_1}{2p_2} + p_2 \times \left(-\frac{p_1^2}{4p_2^2} \right) = \frac{p_1^2}{2p_2} - \frac{p_1^2}{4p_2} = \frac{p_1^2}{4p_2}$$

If we normalize $p_2 = 1$, then profits are 8 in equilibrium. This makes sense because revenue is $\sqrt{8} \times 2\sqrt{8}$ and costs are 8×1 . The consumer's budget set is then:

$$p_1 x_{11} + p_2 x_{21} \leq 24p_2 + \frac{p_1^2}{4p_2}$$

The budget set will bind at the optimum. Solving for x_{11} :

$$\begin{aligned} p_1 x_{11} &= 24p_2 + \frac{p_1^2}{4p_2} - p_2 x_{21} \\ x_{11} &= 24\frac{p_2}{p_1} + \frac{p_1}{4p_2} - \frac{p_2}{p_1} x_{21} \\ x_{11} &= (24 - x_{21}) \frac{p_2}{p_1} + \frac{p_1}{4p_2} \end{aligned}$$

Substituting this into the consumer's objective:

$$\max_{x_{21}} \left[(24 - x_{21}) \frac{p_2}{p_1} + \frac{p_1}{4p_2} \right] x_{21} = 24\frac{p_2}{p_1} x_{21} - \frac{p_2}{p_1} x_{21}^2 + \frac{p_1}{4p_2} x_{21}$$

The first-order condition at an interior solution is:

$$\left[(24 - x_{21}) \frac{p_2}{p_1} + \frac{p_1}{4p_2} \right] - \frac{p_2}{p_1} x_{21} = 0$$

Solving for x_{21} :

$$\begin{aligned} \left[(24 - x_{21}) \frac{p_2}{p_1} + \frac{p_1}{4p_2} \right] &= \frac{p_2}{p_1} x_{21} \\ 24\frac{p_2}{p_1} - \frac{p_2}{p_1} x_{21} + \frac{p_1}{4p_2} &= \frac{p_2}{p_1} x_{21} \\ 24\frac{p_2}{p_1} + \frac{p_1}{4p_2} &= 2\frac{p_2}{p_1} x_{21} \\ 12\frac{p_2}{p_1} + \frac{p_1}{8p_2} &= \frac{p_2}{p_1} x_{21} \\ 12 + \frac{p_1^2}{8p_2^2} &= x_{21} \end{aligned}$$

This makes sense because $x_{21} = 12 + \frac{4 \times 8}{8} = 16$.

Using this in the expression for x_{11} :

$$\begin{aligned} x_{11} &= (24 - x_{21}) \frac{p_2}{p_1} + \frac{p_1}{4p_2} \\ &= \left(24 - \left(12 + \frac{p_1^2}{8p_2^2} \right) \right) \frac{p_2}{p_1} + \frac{p_1}{4p_2} \\ &= \left(12 - \frac{p_1^2}{8p_2^2} \right) \frac{p_2}{p_1} + \frac{p_1}{4p_2} \\ &= 12\frac{p_2}{p_1} - \frac{p_1}{8p_2} + \frac{p_1}{4p_2} \\ &= 12\frac{p_2}{p_1} + \frac{p_1}{8p_2} \end{aligned}$$

Check using prices: $(24 - 16) \frac{1}{2\sqrt{8}} + \frac{2\sqrt{8}}{4} = \frac{\sqrt{8}}{2} + \frac{\sqrt{8}}{2} = \sqrt{8}$